



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

November 2013

Assessment Task 1
Year 11

Mathematics Extension

General Instructions

- Reading Time – 5 Minutes
- Working time – 90 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- Each Section is to be returned in a separate bundle.
- Marks may **NOT** be awarded for messy or badly arranged work.
- All necessary working should be shown in every question if full marks are to be awarded.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 70

Section A (10 marks)

- Answer questions 1-10 on the Multiple Choice answer sheet provided.

Section B (60 Marks)

- For Questions 11-14, start a new answer booklet for each question.

Examiner: *J. Chen*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

SECTION A [10 marks]

Attempt Questions 1-10

Use the multiple-choice answer sheet for Questions 1-10.

Marks

1. Consider the parametric equations $x = t + 2$ and $y = 4t^2$, find the Cartesian equation. **1**
- (A) $y = 4(x + 2)^2$
- (B) $y = 4(x - 2)^2$
- (C) $y = 4\sqrt{x + 2}$
- (D) $y = 4\sqrt{x - 2}$
2. What is the solution to the inequality $\frac{2}{x+1} \geq 1$? **1**
- (A) $-1 < x \leq 1$
- (B) $-1 \leq x \leq 1$
- (C) $x < -1$ and $x \geq 1$
- (D) $x \leq -1$ and $x \geq 1$
3. What is the exact value of $\sin 75^\circ$? **1**
- (A) $\frac{\sqrt{6}-\sqrt{2}}{4}$
- (B) $\frac{\sqrt{2}-\sqrt{6}}{4}$
- (C) $\frac{1-\sqrt{3}}{2\sqrt{2}}$
- (D) $\frac{1+\sqrt{3}}{2\sqrt{2}}$
4. What is the acute angle to the nearest degree between the lines $2x + 3y = 0$ and $y - 3x + 5 = 0$? **1**
- (A) 38°
- (B) 67°
- (C) 74°
- (D) 75°

5. What are the coordinates of the point which divides the interval with end points $(-3, 5)$ and $(7, -5)$ internally in the ratio 2: 3? 1
- (A) $(-23, 25)$
- (B) $(1, 1)$
- (C) $(3, -1)$
- (D) $(-27, -25)$
6. $1 - \frac{\sin x \cos x}{\tan(90^\circ - x)} =$ 1
- (A) 0
- (B) $\sin^2 x$
- (C) $\cos^2 x$
- (D) $\frac{1}{\cos^2 x}$
7. If k is a constant such that $x^3 - kx^2 + kx - 4$ is divisible by $x - k$, then $k =$ 1
- (A) -2
- (B) 0
- (C) 2
- (D) -2 or 2
8. Solve the equation $\frac{1}{\sin \theta + 1} = -1$, where $0^\circ \leq \theta \leq 90^\circ$. 1
- (A) 0°
- (B) 30°
- (C) 90°
- (D) No solutions
9. If $a < 0 < b$, which of the following must always be true? 1
- (A) $a + 1 < b$
- (B) $\frac{1}{a} < \frac{1}{b}$
- (C) $a^2 > b^2$
- (D) $-3a < -2b$

10. A student committee consists of 20 boys and 15 girls. A team of 8 students is selected from the committee to participate in an activity. Find the probability that the team chosen consists of at least 3 boys and at least 3 girls, correct to 3 decimal places.

1

(A) 0.022

(B) 0.275

(C) 0.281

(D) 0.726

End of Multiple Choice Section

SECTION B [60 marks]

Attempt Questions 11-14

Answer each question in a SEPARATE writing booklet.

Question 11 [15 marks] – Use a SEPARATE writing booklet.	Marks
(a) Differentiate the following with respect to x : $\sin(x^\circ)$	2
(b) How many numbers greater than 500 can be formed with the digits 2, 3, 5, 6 and 8 if no digit is repeated?	2
(c) Let $f(x) = 2x^3 + kx^2 - (2k - 1)x - 6$, and $x + 1$ is a factor of $f(x)$. (i) Find the value of k . (ii) Let $g(x) = x^2 - 15x - 16$. Find the quadratic factor of $f(x) - g(x)$.	1 2
(d) Find the gradients of the 2 lines which make angles of 45° with the line whose equation is $3x - 4y + 5 = 0$.	2
(e) If $x = \sqrt{5} \cos 2\theta$ and $y = 3 \cos^2 \theta$, find an equation independent of θ .	2
(f) The equation of the chord of contact from $P(x_0, y_0)$ is $xx_0 = 2a(y + y_0)$. [DO NOT PROVE THIS] Find the coordinates of the point which gives a chord of contact of $x - 3y - 15 = 0$ to the parabola $x^2 = 12y$.	2
(g) Find the exact value of $\sin 15^\circ \cos 15^\circ$.	2

End of Question 11

Question 12 [15 marks] – Use a SEPARATE writing booklet.**Marks**

- (a) $P(2ap, ap^2)$ is a variable point on the parabola $x^2 = 4ay$. The tangent to the parabola at P meets the x -axis at A and the y -axis at B .
- (i) Find the equation of the tangent at P . **1**
- (ii) Find the coordinates of A and B . **1**
- (iii) If M is the midpoint of A and B , describe the locus of M . **2**
- (b) Solve $\sin x + \cos x + 1 = 0$ for $0^\circ \leq x \leq 360^\circ$ by using the substitution $t = \tan \frac{x}{2}$. **3**
- (c) Find the primitives of the following functions with respect to x .
- (i) $y = \sec^2(3x - 2)$ **1**
- (ii) $y = \frac{x}{3x^2 - 5}$ **2**
- (d) A bag contains five cards numbered 1, 2, 3, 4 and 5 respectively. Mary repeats drawing one card at a time randomly from the bag without replacement until the number on the card drawn is 4. Find the total number of ways. **2**
- (e)
- (i) Sketch the curve $y = x \ln x$, showing stationary points and intercepts. **2**
- (ii) Hence, or otherwise, find the value(s) of k for which $x \ln x = k$ has 1 solution. **1**

End of Question 12

Question 13 [15 marks] – Use a SEPARATE writing booklet.**Marks**

(a) If α, β and γ are the roots of $x^3 - 2x^2 + 5x - 4 = 0$, find the value of

(i) $\alpha + \beta + \gamma$ 1

(ii) $\alpha\beta\gamma$ 1

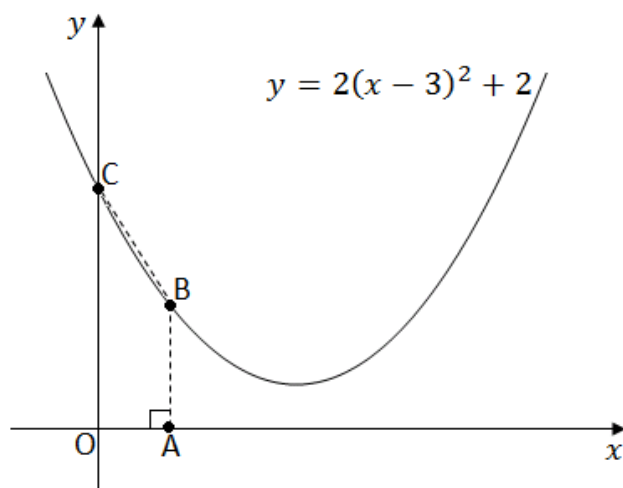
(iii) $\alpha^2 + \beta^2 + \gamma^2$ 2

(iv) $\alpha^3 + \beta^3 + \gamma^3$ 2

(b)

(i) Show that $2x - 3$ is a factor of $8x^3 - 24x^2 + 40x - 33$. 1

(ii) The diagram below shows the graph of $y = 2(x - 3)^2 + 2$. B is a variable point on the graph in the first quadrant. A is the foot of the perpendicular from B to the x -axis. C is the point of intersection of the graph $y = 2(x - 3)^2 + 2$ and the y -axis.



(α) Let $(2m, 0)$ be the coordinates of A . Express the area of the trapezium $OABC$ in terms of m . 2

(β) If the area of the trapezium $OABC$ is 33, find the coordinates of B . 2

(c) Find the general solution of the trigonometric equation $\sin 3\theta = \sin \theta$, express your answer in radians. 4

End of Question 13

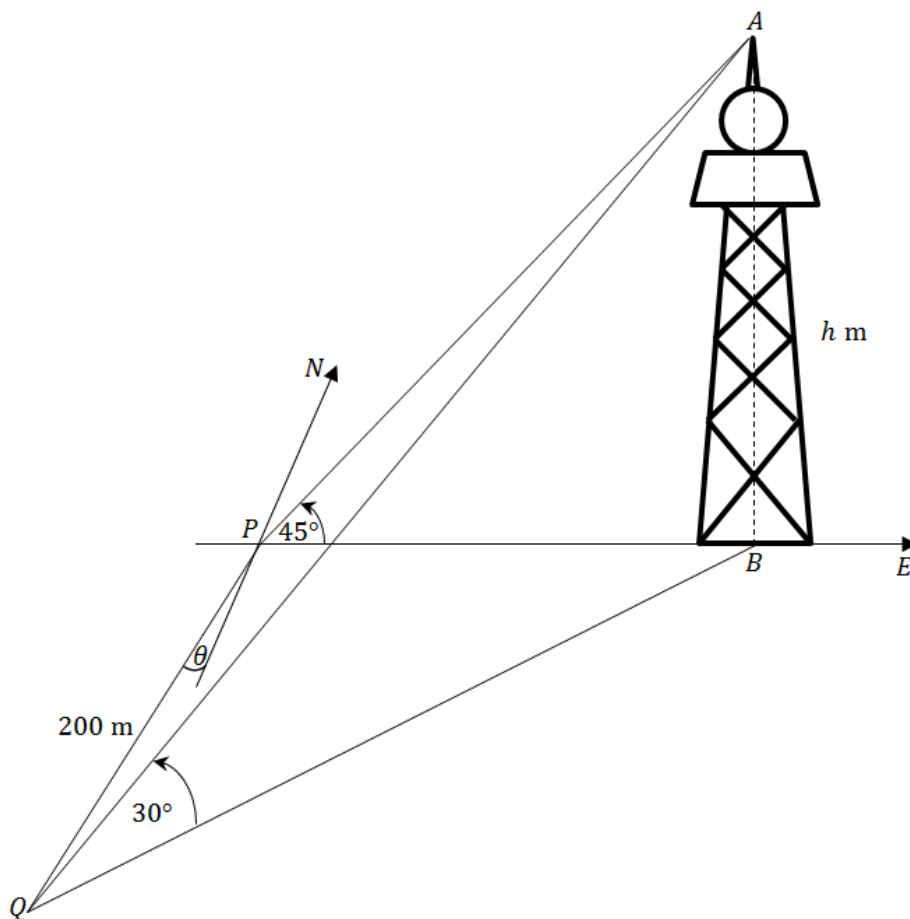
Question 14 [15 marks] – Use a SEPARATE writing booklet.

	Marks
(a)	
(i) Write $2 \sin x - \cos x$ in the form $R \sin(x - \alpha)$, correct α to the nearest degree.	2
(ii) Hence, solve $2 \sin x - \cos x = 1$ for $0 \leq x \leq 360^\circ$.	1
(b) There are 5 different maths classes in year 12. To form a graduate dinner committee of 15 members, 3 representatives are nominated by each class. From the committee, 4 members are randomly selected.	
(i) Find the number of ways if the 4 selected members are nominated by 2 different classes.	2
(ii) Find the probability that the 4 selected members are nominated by at least 3 different classes.	2
(c) For $0 \leq \theta \leq \frac{\pi}{2}$, find the greatest value of	3

$$\frac{\cos^2\left(\frac{\pi}{2} - \theta\right) - 2 \cos^2 \theta}{20}$$

Question 14 continues on next page

- (d) The diagram below shows a tower AB with height h m. The angle of elevation of A from the observers P and Q are 45° and 30° respectively. The bearing of Q from P is $S\theta W$ and the bearing of AB from Q is $N(30^\circ + \theta)E$. The distance between P and Q is 200 m.



- | | | |
|-------|--|---|
| (i) | Show that $h^2 - (200 \sin \theta)h - 20000 = 0$. | 2 |
| (ii) | Show that $\theta = 30^\circ$. | 2 |
| (iii) | Hence, find the value of h . | 1 |

End of Question 14
End of Exam

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Student Number: Answers

Mathematics Extension Task 1 2013

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
correct

Section I: Multiple choice answer sheet.

Completely colour the cell representing your answer. Use black pen.

1. A ☒ B ☐ C ☐ D ☐
2. A ☒ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☒ D ☐
4. A ☐ B ☐ C ☒ D ☐
5. A ☐ B ☒ C ☐ D ☐
6. A ☐ B ☒ C ☐ D ☐
7. A ☐ B ☐ C ☒ D ☐
8. A ☐ B ☐ C ☒ D ☐
9. A ☐ B ☒ C ☐ D ☐
10. A ☐ B ☐ C ☒ D ☐

Q 11.

(a) $y = \sin(x^\circ)$

$$= \sin\left(x \times \frac{\pi}{180}\right)$$

$$y' = \frac{\pi}{180} \cos\left(\frac{\pi}{180}x\right)$$

$$\text{or } \frac{\pi}{180} \cos(x^\circ)$$

(2)

(b) No's > 500 with 2, 3, 5, 6, 8 - no repetitions

3 digits $\boxed{3} \times \underline{4} \times \underline{3} = 36$
 $\uparrow$
 5, 6 or 8

or 4 digits $\boxed{5} \times \boxed{4} \times \boxed{3} \times \boxed{2} = 120$

or 5 digits $\underline{5} \times \underline{4} \times \underline{3} \times \underline{2} \times \underline{1} = 120$

Total ways = 276 ways $\frac{1}{2}$

(2)

(c) $f(x) = 2x^3 + kx^2 - (2k-1)x - 6$

(i) $f(-1) = 0 \Rightarrow 0 = -2 + k + 2k - 1 - 6$

$$3k = 9$$

$$\underline{k = 3}$$

(ii) $g(x) = x^2 - 15x - 16$

Then $f(x) - g(x) = 2x^3 + 3x^2 - 5x - 6 - (x^2 - 15x - 16)$
 $= 2x^3 + 2x^2 + 10x + 10$
 $= 2(x^3 + x^2 + 5x + 5)$

Try $x = -1 \Rightarrow -1 + 1 - 5 + 5 \checkmark \checkmark \therefore (x+1)$ is a factor.
 $x = -5 \Rightarrow -125 + 25 - 25 + 5 \checkmark$

11(c) (ii) (cont)

$$\begin{array}{r} x^2 + 5 \\ x+1 \overline{) x^3 + x^2 + 5x + 5} \\ \underline{x^3 + x^2} \downarrow \\ 0 + 5x + 5 \\ \underline{5x + 5} \\ 0 \end{array}$$

Then $f(x) - g(x) = 2(x+1)(x^2+5)$

$\therefore$ Quadratic factor is x^2+5 ✓

(2)

(d) $\alpha = 45^\circ$

Line $3x - 4y + 5 = 0$

$$\Rightarrow y = \frac{3}{4}x + \frac{5}{4}$$

$$\Rightarrow m_1 = \frac{3}{4}$$

Then let lines which are 45° to given line be in form
 $y = m_2x + b$

$$\therefore \tan 45^\circ = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$1 = \left| \frac{\frac{3}{4} - m_2}{1 + \frac{3}{4} m_2} \right|$$

(2)

$$\Rightarrow \frac{\frac{3}{4} - m_2}{1 + \frac{3}{4} m_2} = 1 \quad \text{or} \quad \frac{\frac{3}{4} - m_2}{1 + \frac{3}{4} m_2} = -1$$

$$\Rightarrow \frac{3}{4} - m_2 = 1 + \frac{3}{4} m_2 \quad \text{or} \quad \frac{3}{4} - m_2 = -1 - \frac{3}{4} m_2$$

$$\Rightarrow \underline{\underline{m_2 = -\frac{1}{7}}}$$

$$\text{or} \quad \Rightarrow \underline{\underline{m_2 = 7}}$$

11(e) $x = \sqrt{5 \cos 2\theta}$ (1) and $y = 3 \cos^2 \theta$ (2)

From (2) $y = 3 \left(\frac{1}{2} (\cos 2\theta + 1) \right)$ ✓

$\Rightarrow y = \frac{3}{2} \cos 2\theta + \frac{3}{2}$ (3) ✓

From (1) $x^2 = 5 \cos 2\theta$

$\Rightarrow \cos 2\theta = \frac{x^2}{5}$ (4) ✓

(2)

Sub (4) into (3) $\Rightarrow y = \frac{3x^2}{10} + \frac{3}{2}$ ✓ or $10y = 3x^2 + 15$

(f) $xx_0 = 2a(y + y_0)$

$a = 3$ $\Rightarrow xx_0 = 6(y + y_0)$ (1) ✓

Now $x - 3y - 15 = 0$

$\Rightarrow x = 3(y + 5)$ ✓

$\times 2 \Rightarrow 2x = 6(y + 5)$ ✓ (2)

(2)

Since (1) $\equiv$ (2) $\Rightarrow x_0 = 2, y_0 = 5$

$\Rightarrow$ Point is $(2, 5)$ ✓

(g) $\sin 15^\circ \cos 15^\circ$

Now $\sin x \cos x = \frac{1}{2} \sin 2x$

$= \frac{1}{2} \sin 30^\circ$ ✓✓

$= \frac{1}{2} \times \frac{1}{2}$

$= \frac{1}{4}$ ✓✓

(2)

$$12) a) i) \quad x^2 = 4ay$$

$$y = \frac{x^2}{4a}$$

$$y' = \frac{x}{2a}$$

$$\text{when } x = 2ap$$

$$m_T = \frac{2ap}{2a}$$

$$= p$$

$$y - y_1 = m(x - x_1)$$

$$y - ap^2 = p(x - 2ap)$$

$$y - ap^2 = px - 2ap^2$$

$$y = px - ap^2$$

$$ii) \quad \text{let } y = 0$$

$$px - ap^2 = 0$$

$$px = ap^2$$

$$x = ap$$

$$\therefore A \text{ is } (ap, 0)$$

$$\text{let } x = 0$$

$$y = -ap^2$$

$$\therefore B \text{ is } (0, -ap^2)$$

$$iii) \quad M \text{ is } \left(\frac{ap+0}{2}, \frac{0+(-ap^2)}{2} \right)$$

$$\left(\frac{ap}{2}, -\frac{ap^2}{2} \right)$$

$$x = \frac{ap}{2}$$

$$p = \frac{2x}{a} \quad \text{--- (1)}$$

$$y = -\frac{ap^2}{2} \quad \text{--- (2)}$$

sub (1) into (2)

$$y = -a \frac{\left(\frac{2x}{a} \right)^2}{2}$$

$$y = -\frac{2x^2}{a}$$

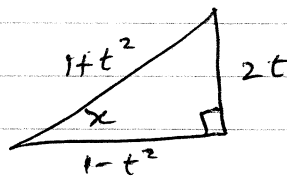
$$x^2 = -\frac{ay}{2}$$

$$x^2 = -4\left(\frac{a}{8}\right)y \quad \text{in the form } x^2 = -4Ay \quad \text{where } A = \frac{a}{8}$$

The locus of M is a parabola
with vertex $(0,0)$, focus $(0, -\frac{a}{8})$

b) $\sin x + \cos x + 1 = 0$

let $t = \tan \frac{x}{2}$



$$0^\circ \leq x \leq 360^\circ$$

$$0^\circ \leq \frac{x}{2} \leq 180^\circ$$

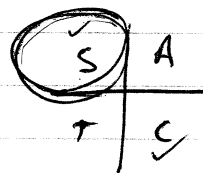
$$\frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} + 1 = 0$$

$$2t + 1 - t^2 + 1 + t^2 = 0$$

$$2t = -2$$

$$t = -1$$

$$\therefore \tan \frac{x}{2} = -1$$



$$\tan \alpha = -1$$

$$\alpha = 45^\circ$$

$$\frac{x}{2} = 135^\circ$$

$$x = 270^\circ$$

must also test $x = 180^\circ$

$$\text{LHS} = \sin 180^\circ + \cos 180^\circ + 1$$

$$= (0) + (-1) + 1$$

$$= 0$$

$$= \text{RHS}$$

$$\therefore x = 180^\circ, 270^\circ$$

c) i) $\int \sec^2(3x-2) dx = \frac{1}{3} \tan(3x-2) + C$

$$\begin{aligned}
 \text{ii)} \quad & \int \frac{x}{3x^2-5} dx \\
 &= \frac{1}{6} \int \frac{6x}{3x^2-5} dx \\
 &= \frac{1}{6} \ln(3x^2-5) + C
 \end{aligned}$$

$$\begin{aligned}
 \text{d)} \quad & \boxed{4} \text{ or } \frac{\boxed{4}}{4 \times 1} \text{ or } \frac{\boxed{4}}{4 \times 3 \times 1} \text{ or } \frac{\boxed{4}}{4 \times 3 \times 2 \times 1} \text{ or } \frac{\boxed{4}}{4 \times 3 \times 2 \times 1 \times 1} \\
 &= 65
 \end{aligned}$$

$$\begin{aligned}
 \text{e) i)} \quad & y = x \ln x \\
 & y' = x \cdot \frac{1}{x} + 1 \cdot \ln x \\
 & y' = 1 + \ln x \\
 & y'' = \frac{1}{x}
 \end{aligned}$$

let $y' = 0$ for stat. points

$$1 + \ln x = 0$$

$$\ln x = -1$$

$$x = e^{-1}$$

$$\text{when } x = \frac{1}{e}$$

$$y'' = \frac{1}{(\frac{1}{e})}$$

$$= e > 0 \quad \cup$$

$$y = \frac{1}{e} \ln\left(\frac{1}{e}\right)$$

$$y = -\frac{1}{e}$$

$\therefore$ Minimum Turning Point at $\left(\frac{1}{e}, -\frac{1}{e}\right)$

Domain: $x > 0$

let $y = 0$

$$0 = x \ln x$$

$$x = 0 \quad \text{or} \quad \ln x = 0$$

but $x > 0$.

$$\underline{x = 1}$$

$$\text{as } x \rightarrow 0^+$$

$$y \rightarrow 0^-$$

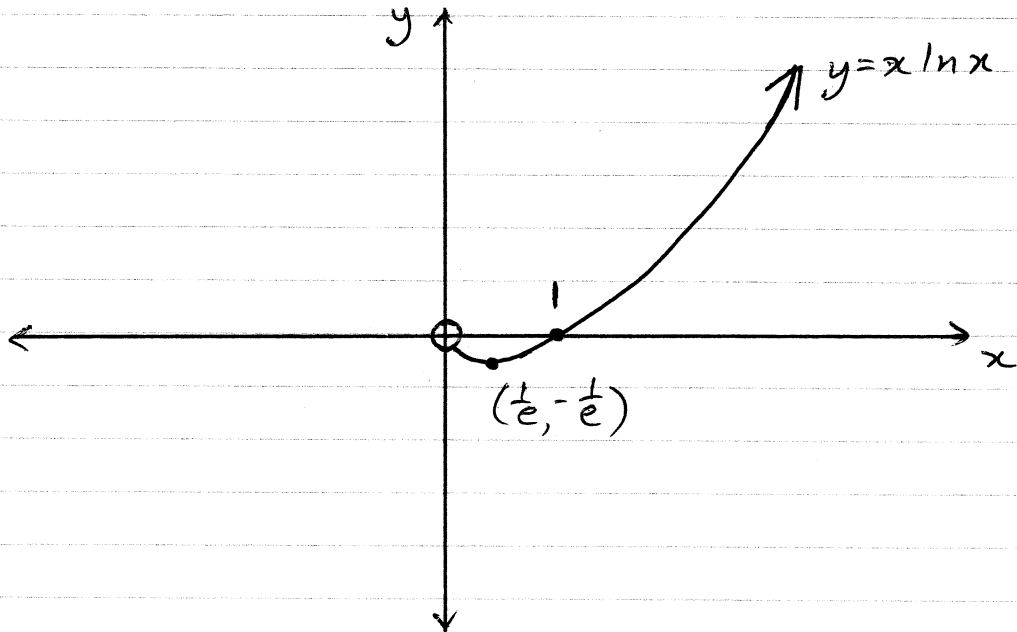
$$\text{as } x \rightarrow +\infty$$

$$y \rightarrow +\infty$$

Note: since $y'' = \frac{1}{x}$

$$\text{and } x > 0$$

$$y'' > 0 \quad \text{CONCAVE UP.}$$



ii) consider where $y = k$ (a horizontal line) intersects the above curve ($y = x \ln x$) in one place.

$$k = -\frac{1}{e}, \quad k \geq 0.$$

QUESTION 13

$$(a) (i) \alpha + \beta + r = 2 \quad 1$$

$$(ii) \alpha\beta r = 4 \quad 1$$

$$(iii) \alpha\beta + \alpha r + \beta r = 5 \quad 2$$

$$(\alpha + \beta + r)^2 = \alpha^2 + \beta^2 + r^2 + 2(\alpha\beta + \alpha r + \beta r)$$

$$\alpha^2 + \beta^2 + r^2 = (\alpha + \beta + r)^2 - 2(\alpha\beta + \alpha r + \beta r)$$

$$= 4 - 10$$

$$= -6$$

$$(iv) \begin{aligned} \alpha^3 + 2\alpha^2 + 5\alpha - 4 &= 0 \\ \beta^3 + 2\beta^2 + 5\beta - 4 &= 0 \\ r^3 + 2r^2 + 5r - 4 &= 0 \end{aligned}$$

$$\alpha^3 + \beta^3 + r^3 + 2(\alpha^2 + \beta^2 + r^2) + 5(\alpha + \beta + r) - 12 = 0$$

$$\alpha^3 + \beta^3 + r^3 + 10 + 10 - 12 = 0.$$

$$\alpha^3 + \beta^3 + r^3 = -10. \quad 2$$

$$(b)(i)(2x-3) \quad \frac{3}{2}.$$

$$\begin{aligned} & 8\left(\frac{3}{2}\right)^3 - 24\left(\frac{3}{2}\right)^2 + 40\left(\frac{3}{2}\right) - 33 \\ &= 27 - 54 + 60 - 33 \\ &= 0 \end{aligned}$$

$$\begin{aligned} (ii) \quad (x) \quad & A(2m, 0) \\ & B(2m, 2(2m-3)^2 + 2) \\ & C(0, 20) \end{aligned}$$

$$\begin{aligned} \text{Area} &= m(2(2m-3)^2 + 2)(20) \\ &= 2m(2m-3)^2 + 22m \\ &= 8m^3 - 24m^2 + 40m. \quad 2 \end{aligned}$$

$$(B) \quad 33 = 8m^3 - 24m^2 + 40m.$$

$$8m^3 - 24m^2 + 40m - 33 = 0.$$

$m = \frac{3}{2}$ is a solution from part (i).

$$B(3, 2). \quad 2$$

$$\begin{aligned}
 (c) \quad \sin 3\theta &= \sin 2\theta \cos \theta + \cos 2\theta \sin \theta \\
 &= 2\sin \theta \cos^2 \theta + \cos^2 \theta \sin \theta - \sin^3 \theta \\
 &= 3\sin \theta (1 - \sin^2 \theta) - \sin^3 \theta \\
 &= 3\sin \theta - 4\sin^3 \theta
 \end{aligned}$$

$$\therefore 3\sin \theta - 4\sin^3 \theta = \sin \theta$$

$$4\sin^3 \theta - 2\sin \theta = 0$$

$$2\sin^3 \theta - \sin \theta = 0$$

$$\sin \theta (2\sin^2 \theta - 1) = 0$$

$$\sin \theta = 0$$

$$\sin \theta = \pm \frac{1}{\sqrt{2}}$$

$$\theta = n\pi$$

$$\theta = n\pi \pm \frac{\pi}{4}$$

$$\text{where } n \in \mathbb{Z}.$$

$$\theta = \frac{\pi n}{2} \pm \frac{\pi}{4}$$

2013 Extension Mathematics Task 1:
Solutions— Question 14

14. (a) (i) Write $2 \sin x - \cos x$ in the form $R \sin(x - \alpha)$, correct α to the nearest degree. 2

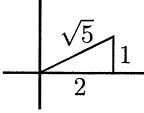
Solution:

$$R = \sqrt{2^2 + 1^2},$$

$$= \sqrt{5}.$$

$$\tan \alpha = \frac{1}{2},$$

$$\alpha \approx 27^\circ.$$

$$\therefore 2 \sin x - \cos x = \sqrt{5} \sin(x - 27^\circ).$$


- (ii) Hence, solve $2 \sin x - \cos x = 1$ for $0^\circ \leq x \leq 360^\circ$. 1

Solution:

$$\sqrt{5} \sin(x - 27^\circ) = 1,$$

$$x - 27^\circ = \sin^{-1} \frac{1}{\sqrt{5}},$$

$$= 27^\circ, (180 - 27)^\circ,$$

$$\therefore x = 54^\circ, 180^\circ.$$

If using the calculated (not rounded) value of α ,
the result is $x = 53^\circ, 180^\circ$.

- (b) There are 5 different maths classes in year 12. To form a graduate dinner committee of 15 members, 3 representatives are nominated by each class. From the committee, 4 members are randomly selected.

- (i) Find the number of ways if the 4 selected members are nominated by 2 different classes. 2

Solution: Ways of choosing classes $= {}^5C_2,$
 $= 10.$

The possibilities are 3 & 1 or 2 & 2 or 1 & 3,
 Ways of choosing 3 & 1 $= 1 \times 3,$
 Ways of choosing 2 & 2 $= {}^3C_2 \times {}^3C_2,$
 $= 9,$
 Ways of choosing 1 & 3 $= 3 \times 1,$
 $\therefore \text{Total ways} = 10(3 + 9 + 3),$
 $= 150.$

- (ii) Find the probability that the 4 selected members are nominated by at least 3 different classes. 2

Solution: Ways of not choosing 2 $= {}^{15}C_4 - 150,$
 $= 1215.$

$$\therefore \text{Probability(at least 3 different)} = \frac{1215}{{}^{15}C_4},$$

$$= \frac{81}{91}.$$

(c) For $0 \leq \theta \leq \frac{\pi}{2}$, find the greatest value of

$$\frac{\cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta}{20}.$$

Solution:
$$\frac{\cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta}{20} = \frac{\sin^2\theta - 2\cos^2\theta}{20}.$$

Clearly, as θ goes from 0 to $\frac{\pi}{2}$, $\sin\theta$ and also $\sin^2\theta$ goes from 0 to 1 and $\cos\theta$ and also $\cos^2\theta$ goes from 1 to 0. Over this domain, $\sin^2\theta$ is monotonic increasing and $\cos^2\theta$ is monotonic decreasing. So the sum, $\sin^2\theta - 2\cos^2\theta$, is monotonic increasing.

$\therefore$ Maximum at $\frac{\pi}{2}$ is $\frac{1}{20}$.

Solution: Alternative method—

$$\begin{aligned}\text{Put } f(\theta) &= \cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta, \\ &= \sin^2\theta - 2\cos^2\theta.\end{aligned}$$

$$\begin{aligned}f'(\theta) &= 2\sin\theta\cos\theta + 4\sin\theta\cos\theta, \\ &= 3\sin 2\theta, \\ &= 0 \text{ when } 2\theta = 0, \pi,\end{aligned}$$

i.e. when $\theta = 0, \frac{\pi}{2}$ we have a stationary point.

$$\text{Testing, } f(0) = 0 - 2 = -2,$$

$$f\left(\frac{\pi}{2}\right) = 1 - 0 = 1.$$

So the maximum is $\frac{1}{20}$ when $\theta = \frac{\pi}{2}$.

Solution: Another alternative—

$$\begin{aligned}\cos^2\left(\frac{\pi}{2} - \theta\right) - 2\cos^2\theta &= \sin^2\theta - 2\cos^2\theta, \\ &= 1 - \cos^2\theta - 2\cos^2\theta, \\ &= 1 - 3\cos^2\theta.\end{aligned}$$

$$\begin{array}{lll} 0 \leq & \theta & \leq \frac{\pi}{2}, \\ \text{but } \cos 0 \geq & \cos \theta & \geq \cos \frac{\pi}{2}, \end{array}$$

$$1 \geq \cos \theta \geq 0,$$

$$1 \geq \cos^2 \theta \geq 0,$$

$$-3 \leq -3\cos^2 \theta \leq 0,$$

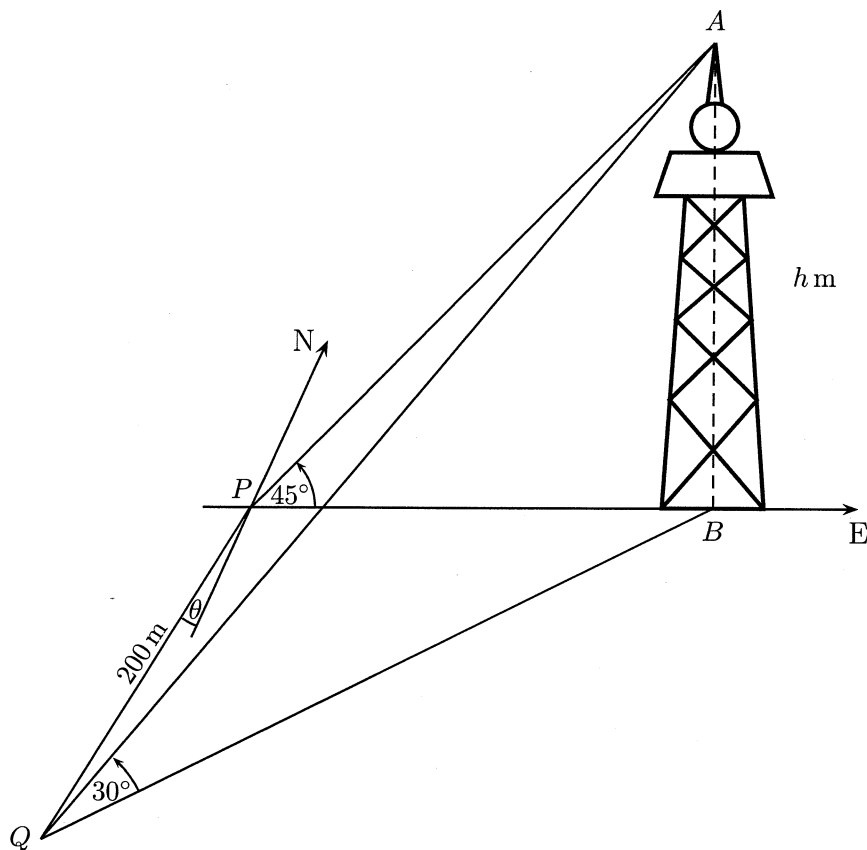
$$-2 \leq 1 - 3\cos^2 \theta \leq 1,$$

$$\frac{-2}{20} \leq \frac{1 - 3\cos^2 \theta}{20} \leq \frac{1}{20},$$

$$-\frac{1}{10} \leq \frac{1 - 3\cos^2 \theta}{20} \leq \frac{1}{20}.$$

i.e. the maximum is $\frac{1}{20}$ when $\theta = \frac{\pi}{2}$.

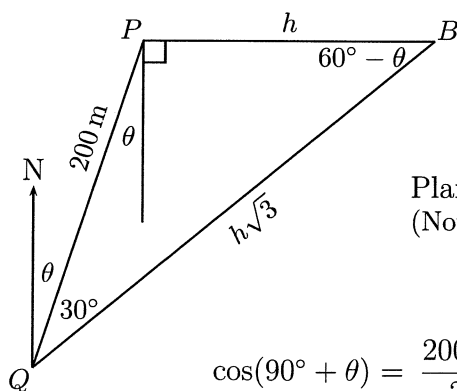
- (d) The diagram below shows a tower AB with height h m. The angles of elevation of A from the observers P and Q are 45° and 30° respectively. The bearing of Q from P is $S\theta W$ and the bearing of AB from Q is $N(30^\circ + \theta)E$. The distance between P and Q is 200 m.



- (i) Show that $h^2 - (200 \sin \theta)h - 20\,000 = 0$.

2

Solution: $\frac{h}{PB} = \tan 45^\circ$, $\frac{h}{QB} = \tan 30^\circ$,
 $\therefore PB = h$, $QB = h\sqrt{3}$.



Plan view
(Not to scale)

$$\cos(90^\circ + \theta) = \frac{200^2 + h^2 - 3h^2}{2 \times 200 \times h},$$

$$-\sin \theta \times 400h = 40\,000 - 2h^2,$$

$$\therefore h^2 - (200 \sin \theta)h - 20\,000 = 0.$$

(ii) Show that $\theta = 30^\circ$.

2

Solution:

$$\frac{k}{\sin 30^\circ} = \frac{k\sqrt{3}}{\sin(90^\circ + \theta)},$$
$$2 \cos \theta = \sqrt{3},$$
$$\cos \theta = \frac{\sqrt{3}}{2},$$
$$\therefore \theta = 30^\circ.$$

(iii) Hence, find the value of h .

1

Solution: Substitute (ii) in (i)—

$$h^2 - 200 \times \frac{1}{2} \times h - 20\,000 = 0,$$

$$h^2 - 100h + 50^2 = 20\,000 + 2500,$$

$$(h - 50)^2 = 22\,500,$$

$$h - 50 = \pm 150,$$

$$h = 200 \text{ or } -100.$$

i.e. h is 200 m (as $h > 0$).